A Common Recovery Transition for Classical and Quantum Information in Random Codes

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Classical reading and quantum recovery impose different demands on a code. For Pauli noise and any exact unitary 2-design, we match a converse for optimal complete-message success to achievable entanglement fidelity. With fixed i.i.d. single-qubit noise, Shannon entropy $0 < H < 1$, and positive self-information variance, both optima follow one Gaussian curve in the positive-rate critical window, in mean and with high probability for the same encoder. Their ensemble-mean coding sizes agree through second order.

Classical reading tolerates logical phase errors that spoil quantum coherence. A quantum code must preserve both logical populations and coherences [1]. Recovery from two complementary bases connects these requirements [2]. In Haar-random Hayden–Preskill dynamics [3], Nakata, Matsuura, and Koashi found a common classical and quantum recovery threshold [4]. A common threshold identifies where reliable recovery becomes possible; it leaves open how the two optimal performances compare at fixed nonzero error throughout the transition.

Here a converse for optimal classical decoding closes this gap for Pauli noise. It matches quantum achievability in a finite-size information-spectrum bracket valid for every exact unitary 2-design. The two tasks use the *same* random encoder: the classical receiver guesses the entire uniformly distributed logical-basis message, and the quantum receiver restores entanglement with a reference. Each knows the realized encoder and optimizes its decoder separately. For a fixed i.i.d. single-qubit error distribution with entropy $0 < H < 1$ and positive self-information variance, the bracket fixes a common Gaussian curve in the positive-rate critical window and common first- and second-order terms in the ensemble-mean coding sizes. It also controls both performances jointly for the same sampled code. Classical phase-error tolerance produces a finite-size advantage without changing these leading terms.

The hashing rate $1 - H(\boldsymbol{p})$ is an established quantum-coding benchmark [5, 6]. Random coding and one-shot decoupling provide quantum achievability using Haar randomness and unitary designs [7–9]. Klesse's bound yields truncated-error-spectrum achievability and a probability bound over encoders [10]. Finite-resource quantum bounds [11] and Clifford analyses [12] quantify second-order rates. Erasure-code recovery [13] and Haar-code spectra [14] exhibit fluctuation-broadened coding transitions; approximate quantum Hamming bounds [15] and stabilizer converses [16] provide complementary constraints. In the trace-distance formulation of Ref. [4], the recovery bound has square-root dependence on classical errors. The matching converse here bounds the classical optimum directly, allowing quantum achievability to determine both optimal performances at fixed nonzero error in the regular critical window.

One encoder, two tasks.—Let $d = 2^n$ and $M = 2^k \le d$, with integer $k \ge 1$. Embed the logical space by $V_0|j\rangle = |j\rangle|0\rangle^{\otimes(n-k)}$ and encode with $V = UV_0$, where U is drawn from an exact unitary 2-design. Haar and uniform Clifford encoders are examples [17]. All codewords are orthogonal columns of the same U, retaining their correlations. A fixed Pauli channel acts after encoding:

$$\mathcal{N}_p(\rho) = \sum_{P\in\{I,X,Y,Z\}^{\otimes n}} p_P P\rho P^\dagger, \quad s(P) = -\log_2 p_P. \tag{1}$$

The distribution is independent of U and may include correlated errors. Define $\mathcal{F}(t) = \mathbb{P}_{P\sim p}[s(P) < t]$. All logarithms are base two.

For $\rho_j = \mathcal{N}_p(V|j\rangle\langle j|V^\dagger)$, the optimal classical success probability is

$$P_{\mathrm{c}}^*(V) = \max_{\{E_j\}} \frac{1}{M} \sum_j \mathrm{tr}(E_j\rho_j). \tag{2}$$

Let $|\Phi_M\rangle = M^{-1/2}\sum_j |j\rangle|j\rangle$ and $\mathcal{V}(\rho) = V\rho V^\dagger$. The optimal entanglement fidelity [18] is

$$F_{\mathrm{q}}^*(V) = \max_{\mathcal{D}} \langle\Phi_M|[\mathrm{id}\otimes(\mathcal{D}\mathcal{N}_p\mathcal{V})](\Phi_M)|\Phi_M\rangle, \tag{3}$$

where $\mathcal{D}$ is any quantum channel back to the logical space. Fidelity here is the overlap, not its square root. Both optimizations precede averaging over U. These are the operational tasks underlying classical and quantum conditional min-entropy [19]. Encoding and decoding are ideal; neither a noise record nor access to the environment is supplied.

Bit errors and phase errors.—Clifford encoders make the distinction concrete through the stabilizer formalism [20]. Conjugating a physical Pauli through the known inverse $U^\dagger$ gives logical bit-flip and phase labels (x, z) and an auxiliary bit-flip syndrome s. Let $w(s, x, z)$ sum the probabilities of all errors with those labels, including auxiliary phase degeneracy. The orthogonal syndrome sectors give

$$\begin{aligned} F_{\mathrm{q}}^* &= \sum_s \max_{x,z} w(s,x,z), \\ P_{\mathrm{c}}^* &= \sum_s \max_x \sum_z w(s,x,z). \end{aligned} \tag{4}$$

Classical decoding gains by pooling all phase labels for the same bit flip; quantum recovery corrects the most likely logical Pauli in each sector [13, 16]. This distinction explains the finite-size classical advantage. The classical output states commute, so inverse encoding, basis measurement, and classical maximum-likelihood processing attain P_c^*. Evaluating the required probabilities can be computationally costly. The Supplemental Material (SM) [21] proves both optimality statements. The common bracket below extends the comparison to every exact unitary 2-design, including non-Clifford encoders.

A common finite-size bracket.—The classical converse and quantum achievable bound are controlled by the same error spectrum. As in information-spectrum converses [22, 23], its distribution retains the fluctuations that a mean entropy alone cannot describe.

Theorem 1. *For any distribution in Eq. (1), any exact unitary 2-design encoder, and every $a > 0$,*

$$\mathcal{F}(n-k-a) - 8\,2^{-a} \le \mathbb{E}F_q^* \le \mathbb{E}P_c^* \le \mathcal{F}(n-k+2a) + \frac{5}{4}\,2^{-a} + \frac{2^a}{4M}. \tag{5}$$

Moreover, $F_q^*(V) \le P_c^*(V)$ for every encoder. With $F_- = \mathcal{F}(n-k-a)$ and $F_+ = \mathcal{F}(n-k+2a)$, for any $\eta > 0$,

$$\mathbb{P}_U[F_q^* < F_- - \eta \text{ or } P_c^* > F_+ + \eta] \le \frac{37\,2^{-a}}{4\eta} + \frac{2^a}{4M\eta}. \tag{6}$$

The scale $n-k$ has a familiar coding interpretation. For i.i.d. errors with single-qubit Shannon entropy H, there are roughly 2^{nH} typical errors, each mapping a 2^k-dimensional code into the 2^n-dimensional physical space. Comparing $2^k 2^{nH}$ with 2^n suggests the hashing rate $R = k/n = 1-H$. The theorem turns this counting scale into bounds that control the actual overlaps and degeneracies of the error images. Its classical converse limits the gain from pooling phase errors: that gain cannot shift the asymptotic complete-message threshold. The spectral mass near the coding budget then sets the transition profile, as quantified below.

Proof.—For a recovered logical channel with Kraus operators A_ℓ,

$$F_e = \frac{1}{M^2}\sum_\ell |\operatorname{tr} A_\ell|^2 \le \frac{1}{M}\sum_{j,\ell} |(A_\ell)_{jj}|^2 \le P_c^*. \tag{7}$$

This uses the ensemble-fidelity bound of Ref. [18]: the middle expression is the success of that recovery followed by logical-basis measurement, proving the pointwise ordering.

For the lower bound, take $\mathcal{G} = \{s < n-k-a\}$, $L = |\mathcal{G}| < d2^{-a}/M$, and define

$$S = \sum_{P\in\mathcal{G}} PVV^\dagger P^\dagger, \qquad R_P = V^\dagger P^\dagger S^{-1/2}. \tag{8}$$

The inverse is on the support of S. The R_P obey $\sum_P R_P^\dagger R_P = \Pi_{\mathrm{supp}S}$; completing them on its kernel gives a trace-preserving square-root recovery [24], using the structure of pretty-good measurements [25]. For $v_{jP} = PV|j\rangle$ and $a_{jP} = \langle v_{jP}|S^{-1/2}|v_{jP}\rangle$, Hayashi–Nagaoka [23] gives

$$1 - a_{jP}^2 \le 4 \sum_{(i,Q)\neq(j,P)} |\langle v_{jP}|v_{iQ}\rangle|^2 =: 4C_{jP}. \tag{9}$$

Also $0 \le a_{jP} \le 1$. Keeping the terms where the actual error matches the recovery Kraus index in the fidelity sum yields

$$F_e \ge \sum_{P\in\mathcal{G}} p_P \left(\frac{1}{M}\sum_j a_{jP}\right)^2 \ge p(\mathcal{G}) - 8X,$$
$$X = \sum_{P\in\mathcal{G}} \frac{p_P}{M}\sum_j C_{jP}. \tag{10}$$

The label P in Eq. (8) is an internal Kraus index, not a known noise realization. Exact unitary second moments and orthogonality of the code columns give

$$\mathbb{E}C_{jP} = \frac{(L-1)(dM-1)}{d^2-1} \le \frac{ML}{d}, \tag{11}$$

so $\mathbb{E}X \le 2^{-a}$. This proves the lower bound, including nonuniform error probabilities. The empty $\mathcal{G}$ case is trivial.

For the upper bound, every POVM and every $c > 0$ satisfy

$$P_{\mathrm{succ}} \le \frac{cd}{M} + \frac{1}{M}\sum_j \operatorname{tr}(\rho_j - cI)_+. \tag{12}$$

Split each output at error probability c' into a head and a tail B_j. The head contributes at most its mass, while $(\lambda - c)_+ \le \lambda^2/(4c)$ bounds the tail. Each code column is a state 2-design, and Pauli orthogonality gives

$$\mathbb{E}\operatorname{tr} B_j^2 = \frac{dv_t + m_t^2}{d+1} \le c' + \frac{1}{d}, \tag{13}$$

where $m_t = \sum_{p_P \le c'} p_P$ and $v_t = \sum_{p_P \le c'} p_P^2 \le c' m_t$. Putting $\gamma = cd$ and $c' = c2^{-a}$ gives

$$\mathbb{E}P_c^* \le \frac{\gamma}{M} + \frac{1}{4\gamma} + \frac{2^{-a}}{4} + \mathcal{F}(n - \log_2\gamma + a). \tag{14}$$

Choose $\gamma = M2^{-a}$. The same pointwise constructions control the positive deviations from F_- and F_+; Markov's inequality gives Eq. (6). Full details appear in the SM [21].

Error fluctuations and the shared critical curve.—Error self-information determines both the width and the shape of the shared transition. For independent identical errors, $p = \boldsymbol{p}^{\otimes n}$, write $H = \mathbb{E}[-\log_2 p_\alpha]$ and $\sigma^2 =$

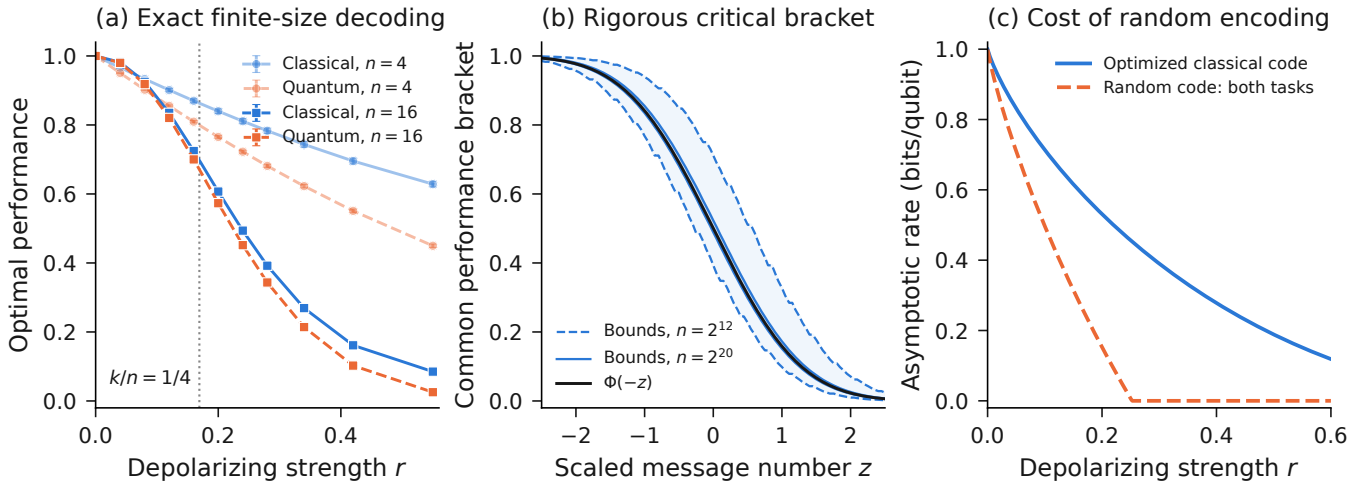


FIG. 1. Two optimal tasks and one asymptotic transition. (a) Exact classical success and entanglement fidelity for uniform Clifford encoders with $k/n = 1/4$; 64 encoders per size, reused across noise strengths. Markers show encoder means and one-standard-error bars; lines join points. The vertical line solves $H(1-3r/4, r/4, r/4, r/4) = 3/4$. At complete depolarization ($r = 1$, outside the plotted range), the no-information baselines are $P_c^* = 1/M$ and $F_q^* = 1/M^2$; for $n = 4, k = 1$ they are $1/2$ and $1/4$. (b) Rigorous bounds in Eq. (5) for $r = 0.1$, $a = 2\log_2 n$, and k chosen by Eq. (15), compared with $\Phi(-z)$. These are binomial-CDF evaluations, not large-system decoding simulations. (c) The common random-code rate $[1-H]_+$ compared with the optimized classical capacity $1 - h_2(r/2)$ of the same depolarizing channel, where h_2 is the binary entropy.

$\mathrm{Var}[-\log_2 p_\alpha]$ for one qubit. The total self-information $s(P)$ is a sum of n independent terms, with mean nH and standard deviation $\sqrt{n}\,\sigma$. For $0 < H < 1$ and $\sigma > 0$, moving the coding budget $n-k$ through this distribution produces a window of width $O(\sqrt{n})$ in k, or $O(n^{-1/2})$ in rate. The central limit theorem fixes its Gaussian shape. Fluctuation analysis underlies second-order channel coding [26, 27] and related Gaussian expansions in quantum hypothesis testing [28, 29]. The matching sides of Eq. (5) make this error-spectrum shape the common limit of both optimized recovery performances.

At fixed positive rate $R = k/n$, the bracket gives exponential approach to one for $R < 1 - H$, and exponential decay to zero for $R > \max(1-H, 0)$. More precisely, fix $\boldsymbol{p}$ with $0 < H < 1$ and $\sigma > 0$, and let

$$k_n = n(1-H) + \sqrt{n}\,\sigma z + O(1) \tag{15}$$

for fixed z. Berry–Esseen [30], applied to the two CDFs in Eq. (5) with $a = 2\log_2 n$, yields

$$\boxed{\begin{aligned} \mathbb{E}F_q^* &= \Phi(-z) + O\left(\frac{\log n}{\sqrt{n}}\right), \\ \mathbb{E}P_c^* &= \Phi(-z) + O\left(\frac{\log n}{\sqrt{n}}\right). \end{aligned}} \tag{16}$$

Here Φ is the normal CDF. Equation (6) with $\eta = n^{-1/2}$ shows that both quantities follow this curve for the same encoder outside an exceptional set of probability at most $37/(4n^{3/2}) + n^{5/2}/(4M_n)$. The constants are uniform over the choice of exact 2-design. For fixed $0 < \epsilon < 1$ and sufficiently large n, define the largest logical sizes satisfying the ensemble-mean error constraint:

$$\begin{aligned} k_{\epsilon,n}^{\mathrm{c}} &= \max\{k : \mathbb{E}_U P_c^*(V_{n,k}) \geq 1-\epsilon\}, \\ k_{\epsilon,n}^{\mathrm{q}} &= \max\{k : \mathbb{E}_U F_q^*(V_{n,k}) \geq 1-\epsilon\}. \end{aligned} \tag{17}$$

Here $k \in \{1, \ldots, n\}$ and the exact 2-design ensemble is prescribed for each (n, k); only the decoders are optimized for each realization. Then

$$\begin{aligned} k_{\epsilon,n}^{t} &= n(1-H) + \sqrt{n}\,\sigma\Phi^{-1}(\epsilon) + O(\log n), \\ & t \in \{\mathrm{c}, \mathrm{q}\}. \end{aligned} \tag{18}$$

The two sizes share the first and second-order terms; their finite-size values need not coincide. Fixed message dimension, zero surprisal variance, and the endpoint $H = 1$ are outside this Gaussian statement; Eq. (5) remains valid there.

Finite-size decoding.—Exact Clifford decoding isolates the classical advantage from combining logical phase errors. Uniform encoders sampled with the Bravyi–Maslov construction [31] give both optima in Eq. (4) for $n = 4, 8, 12, 16$ through an exact binary Fourier sum over syndrome and logical labels. Figure 1(a) displays this advantage at the smallest and largest sizes. Direct Pauli enumeration and primal–dual semidefinite optimization at $n \leq 3$ independently check the optimality formulas and explicit recovery. Methods, all sizes, seeds, and raw encoder values accompany the SM. These samples test the finite-size decoder mechanism; the spectral bounds establish the shared asymptotic curve [Fig. 1(b)].

Reliable decoding and retained information.—The common curve quantifies recovery of the entire encoded mes-

sage. A vanishing P_c^* can coexist with substantial partial information: a receiver who learns half of a uniform k-bit message and nothing about the rest has success $2^{-k/2}$, yet retains $k/2$ bits. Thus the complete-message converse leaves open how much accessible information or which smaller logical subsystems survive.

The comparison with channel capacity locates the cost at the encoder. For $\mathcal{N}_r(\rho) = (1-r)\rho + rI/2$, classical product-state coding achieves capacity $1 - h_2(r/2)$ [32]. At $r = 0.1$ it is 0.714 bits per qubit, whereas the common random-code rate is 0.497. The latter vanishes at $r_c = 0.252386\ldots$, where classical capacity remains 0.453 [Fig. 1(c)]. Codes tailored to classical messages therefore support reliable communication where these random codebooks have vanishing complete-message success at every positive rate. This is a Pauli-noise counterpart of the classical encoding cost identified in Hayden–Preskill dynamics [4]: ignoring coherence at decoding does not reproduce the advantage of tailoring the encoding to classical messages.

Random-state contraction [33] and coherent information [14] probe related aspects of noisy encodings. Liu *et al.* bound accessible information for a full state-2-design alphabet [34]. The conditional alphabet here consists of the M orthogonal columns at fixed, known U, and its decoder is optimized before averaging over U. Extending the joint bracket to approximate designs [9] and noisy encoding remains open and would connect this comparison to low-depth circuit codes [12, 13]. In the regular i.i.d. window for exact designs, the matching classical converse fixes a common optimal critical curve and second-order coding sizes: the two recovery tasks are governed by the same error fluctuations even though their finite-size decoders differ.

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Data availability.—The source code, raw encoder samples, numerical verification records, and figure data are retained in the project research repository. No public archive identifier has been assigned.

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